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**B.E - (Computer) (Sem-VII)(Revised Course 2019-2020)**

**EXAMINATION MAY 2024**

**CE 723 Data Analytics (Professional Elective)**

**[Time: 3:00 Hours]**

**[Max. Marks: 100]**

- Instructions:** 1) Assume suitable data  
2) Answer any 02 questions from PART-A and PART-B each and Attempt 01 question from PART-C

**PART – A**

**Q.1** a) Apply Multiple linear regression for the following dataset 7

| X1 | X2 | Y   |
|----|----|-----|
| 60 | 60 | 140 |
| 62 | 62 | 155 |
| 67 | 67 | 159 |
| 70 | 70 | 179 |
| 71 | 71 | 192 |
| 72 | 72 | 200 |
| 75 | 75 | 212 |
| 78 | 78 | 215 |

b) Blood Pressure (BP) of 8 patients before and after are recorded. Is there any significant difference between BP readings before and after? Analyze. Use  $\alpha = 0.05$  and t value=2.36 7

|        |     |     |     |     |     |     |     |     |
|--------|-----|-----|-----|-----|-----|-----|-----|-----|
| Before | 181 | 201 | 231 | 241 | 171 | 191 | 201 | 166 |
| After  | 141 | 146 | 151 | 156 | 121 | 131 | 141 | 131 |

c) A survey of cars was conducted in 2011 and determined that 60% of car owners have only 1 car, 28% have two cars and 12% have 3 or more cars. Supposing that you have decided to conduct your own survey and have collected the data below, determine whether your data supports the results of the study. Use a significance level of 0.05(5%). Also given that out of 129 car owners, 73 have one car and 38 have 2 cars. Critical value = 5.99. 6

- Q.2** a) A sample size of 16 people is taken. Suppose a claim is made that the average number of days a person spends a vacation is more than or equal to 5 days based on sample of 16 people whose mean came out to be 9 days. Sample standard deviation is found to be 3 days at a level of significance of 0.05 and t value for right-tailed test comes out to be 1.753. Calculate the value of t and check null hypothesis. **6**

- b) The marks of boys and girls are as follows: **6**  
 Boys: 12, 14, 10, 8, 16, 5, 3, 9, 11  
 Girls: 21, 18, 14, 20, 11, 19, 8, 12, 13, 15  
 Is there any significant difference between the marks of girls and boys? Assume  $\alpha=0.05$ , t-value (critical) = 2.11.

- c) To assess the significance of possible variation in a certain test between the convent schools of a city, a common test was given to a number of students taken at random from the fifth class of 3 schools concerned, the results are given below.  $F_{0.05}=3.89$  **8**

| A  | B  | C  |
|----|----|----|
| 9  | 13 | 14 |
| 11 | 12 | 13 |
| 13 | 10 | 17 |
| 9  | 15 | 7  |
| 8  | 5  | 9  |

Make analysis of variance for the given data.

- Q.3** a) In an anti-malaria campaign in a certain area quinine was administered to 812 people out of a total of 3248. The number of fever cases is shown below.  $X^2_{0.05}=3.84$ . Find out whether there is relationship between treatment and fever. **5**

| Treatment     | Forever(A) | No Fever(a) |
|---------------|------------|-------------|
| Quinine(B)    | 141        | 31          |
| No Quinine(b) | 61         | 21          |

- b) Write short notes on logistic regression, ANOVA, Hypothesis test and Probability of error. **4**
- c) Find the value of t-test value for the following set of values: 7, 2, 9, 8 and 1, 2, 3, 4 **4**
- d) Internal test marks of students are as follows: 10.5, 9, 7, 12, 8.5, 6.5, 8, 11, 9.9 **7**  
 Mean population score is 12 and standard deviation is 1.80. Is the mean value for student significantly differed from the mean population value? Let alpha value is 0.05, so corresponding t value is 2.262.

**PART – B**

- Q.4** a) Explain different types of NoSql database. **5**
- b) Explain different ways of reading data in R. **5**
- c) Differentiate between structured and unstructured data. **5**
- d) Explain how a decision tree model is built using R programming language. Also show commands to plot a tree. **5**
- Q.5** a) Explain each phases of data analytic life cycle. **7**
- b) Compare MySql with NoSql database. **6**
- c) Explain the different data types supported in R with the help of examples. **7**
- Q.6** a) Create a data frame named "Student" in R programming languages with three attribute (Roll No, Name, Branch). Also display **8**
- i) All the elements in the first rows
- ii) All the elements of first column
- iii) Element at 2nd row 3rd column
- b) Explain four different ways of creating matrix in R programming language using examples. **5**
- c) Explain all the control statements available in R programming Language with the help of an example. **7**

**Q.7**

**PART – C**

- a) The 3 samples below have been obtained from normal populations with equal variances. Test the hypothesis that the sample means are equal.  $F_{0.05}=3.68$  **8**

| A  | B  | C  |
|----|----|----|
| 8  | 8  | 17 |
| 10 | 6  | 10 |
| 7  | 11 | 12 |
| 14 | 8  | 12 |
| 11 | 8  | 15 |
| 16 | 13 | 12 |

b) Discuss in detail Chi<sup>2</sup> test with detail example. 2

c) The Internal Test 1 marks of B.E Computer Boys and Girls are as follows: 5

Boys: 12, 13, 14, 10, 18, 16, 5, 3, 9, 4, 11;

Girls: 21, 18, 14, 20, 11, 19, 8, 12, 13, 4, 15.

Is there any significant difference between marks of girls and boys? Analyze by considering  $\alpha = 0.05$  and t-value = 2.11

d) Write short notes on Significance level, Directed hypothesis, Probability of error, Hypothesis test. 5

**Q.8** a) Explain database system structure with a neat diagram. 7

b) Explain at least three data visualization function in R programming language with the help of an example. 6

c) Explain different types of database languages. 7